

NETWORK TROUBLESHOOTING, MANAGEMENT , MOINTORING AND OPTIMIZATION

1. Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

```
Autoconfiguration Enabled . . . . . : Yes
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 42-23-38-AD-47-A3
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . . : Yes

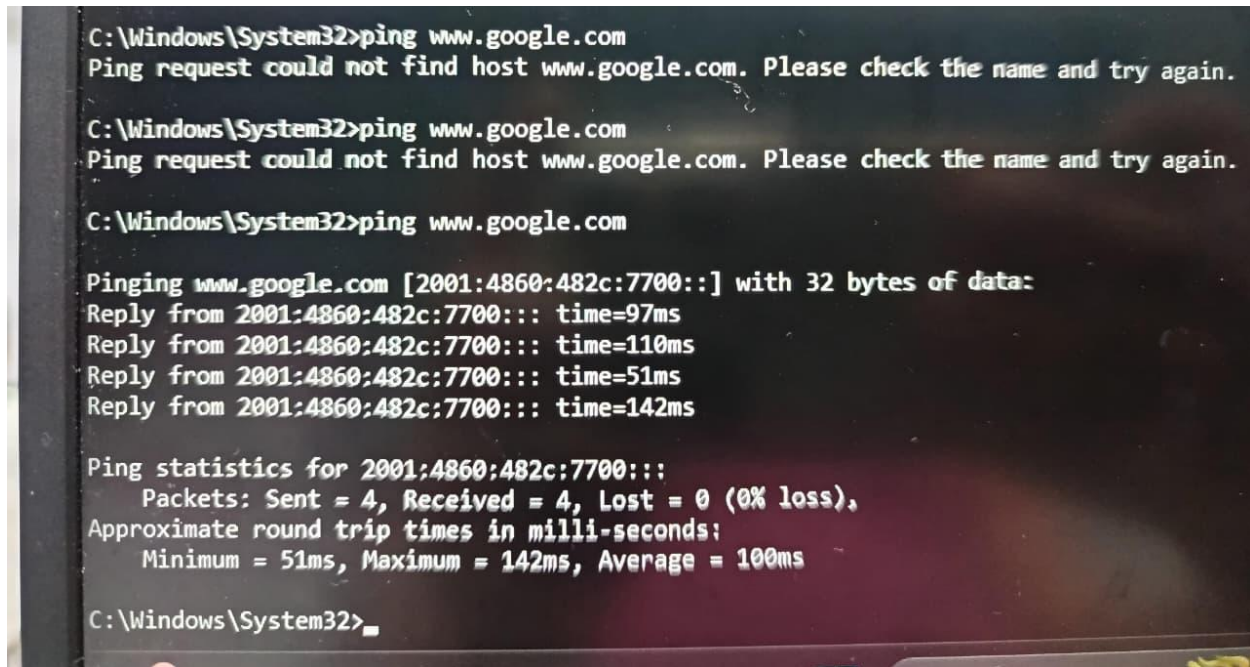
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : 
Description . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : 4C-23-38-AD-47-A3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
IPv6 Address. . . . . : 2402:8100:2682:389a:cfc6:b6e6:85e1:3c7c(Preferred)
IPv6 Address. . . . . : fd80:111:100(Preferrred)
IPv6 Address. . . . . : fd80:111:101(Preferrred)
IPv6 Address. . . . . : 2402:8100:2682:389a:4455:e954:c912:4eeb(Preferrred)
Temporary IPv6 Address. . . . : fe80::7001:1050:a6b7:a104%14(Preferrred)
Link-local IPv6 Address . . . . : fe80::247a:20ff:fe1f:1ec1%14
IPv4 Address. . . . . : 10.111.94.62(Preferrred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, July 28, 2026 7:13:54 PM
Lease Expires . . . . . : Tuesday, July 28, 2026 8:13:53 PM
Default Gateway . . . . . : fe80::247a:20ff:fe1f:1ec1%14
                          10.111.94.200
DHCP Server . . . . . : 10.111.94.200
DHCPv6 IAID . . . . . : 139207480
DHCPv6 Client DUID. . . . . : 00-01-00-01-2F-94-43-B6-BC-FC-E7-8F-3A-A0
DNS Servers . . . . . : 10.111.94.200
                          2402:8100:2682:389a::73
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 4C-23-38-AD-47-A4
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes

C:\Windows\System32>"%UserProfile%"
```

Parameter	Your Value
IPv4 Address	10.111.94.62
Subnet Mask	255.255.255.0
Default Gateway	10.111.94.200
DNS Server(s)	10.111.94.200

2. Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.



```
C:\Windows\System32>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Windows\System32>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Windows\System32>ping www.google.com

Pinging www.google.com [2001:4860:482c:7700::] with 32 bytes of data:
Reply from 2001:4860:482c:7700::: time=97ms
Reply from 2001:4860:482c:7700::: time=110ms
Reply from 2001:4860:482c:7700::: time=51ms
Reply from 2001:4860:482c:7700::: time=142ms

Ping statistics for 2001:4860:482c:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 51ms, Maximum = 142ms, Average = 100ms

C:\Windows\System32>
```

Short Note

- Disconnected: Ping failed because there was no network connection.
- Reconnected: Ping succeeded, showing the computer could communicate with Google's server. This means the Internet connection was restored

3. Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Steps

1. Press Ctrl + Shift + Esc to open Task Manager.
2. Click the Processes tab.
3. Play a YouTube video for 5 minutes.

4. Observe the Network column.

Example Result

Process	Peak Network Usage
Google Chrome	2.9 Mbps

**4. Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.**

Steps

1. Install and open Wireshark.
2. Start capturing on your Wi-Fi or Ethernet interface.
3. Browse Flipkart or Instagram Web for 30 seconds.
4. Stop the capture.
5. In the filter bar, type:
 - http (for HTTP traffic)
 - tls (recommended for HTTPS traffic)

Three Types of Information You Can See

Information	Description
Source IP Address	IP address of the sender.
Destination IP Address	IP address of the receiver/server.
Protocol Details	HTTP request, TLS handshake, packet size, etc.

5. Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first

Home Network Speed & Reliability Checklist

- Restart the modem and Wi-Fi router.
- Move closer to the Wi-Fi router or use a Wi-Fi extender.
- Use an Ethernet cable instead of Wi-Fi for gaming.
- Disconnect unused devices from the network.
- Close background downloads, updates, and streaming apps.
- Update the router firmware.
- Restart the gaming PC or console before playing.

Suggestion I Would Try First

Use an Ethernet cable instead of Wi-Fi.

Reason: A wired connection provides a more stable connection, lower latency (ping), and fewer interruptions during online gaming.